Università degli Studi di Salerno

Tesi di Laurea Magistrale in Internet of Things

Efficient Visual Speech Recognition using Parameter-Efficient Fine-Tuning of Frozen Self-Supervised Backbones

Frozen AV-HuBERT with LoRA adapters, evaluated on MuAViC (Arabic, German, Russian)

Candidato: Zakarya Boudraf 0522501649
Relatori: Prof. Fabio Narducci, Dott. Benedetto Simone

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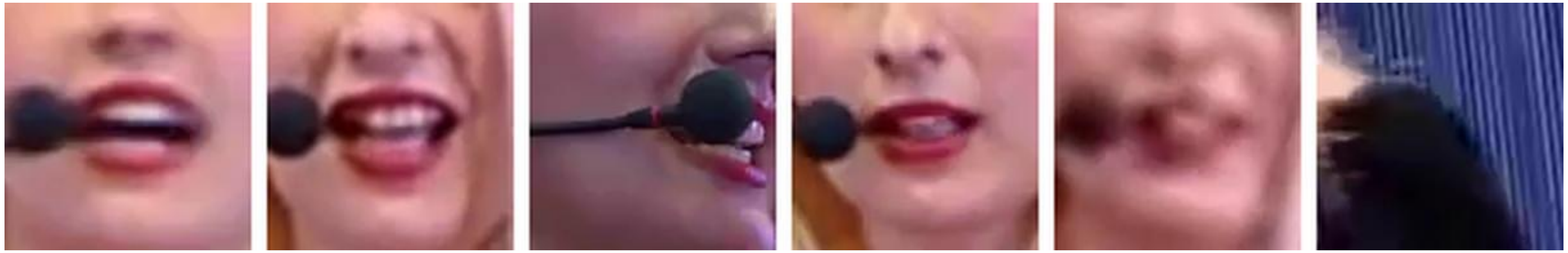
The problem: recognising speech from silent video

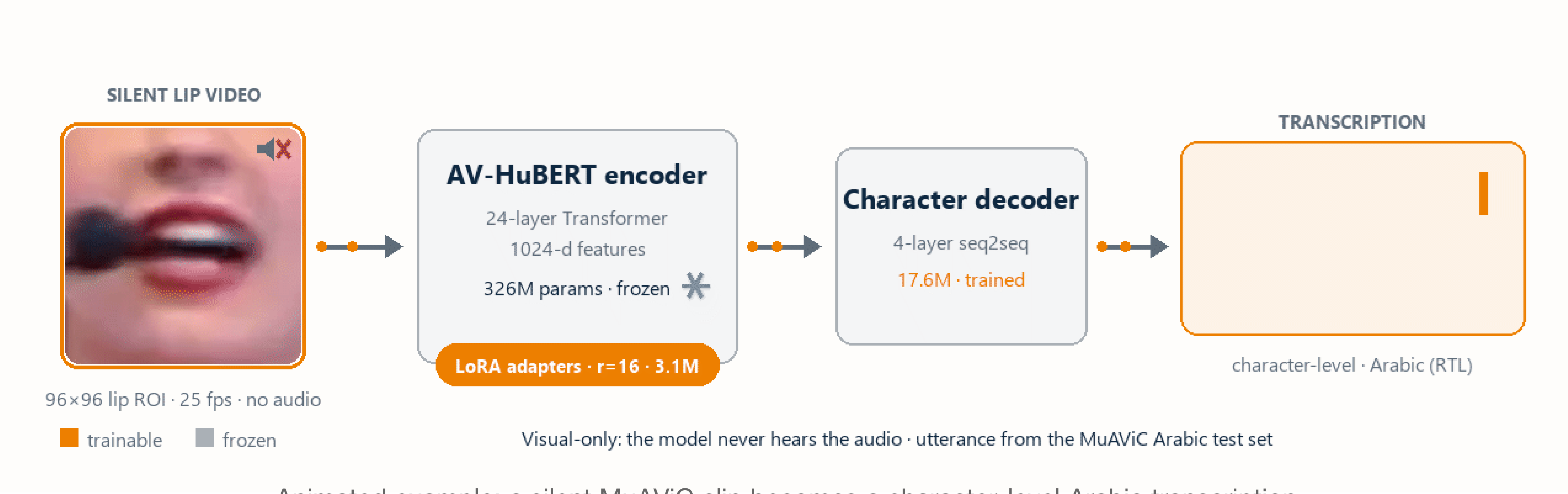
- **Visual speech recognition (VSR):** transcribe what is said from lip movements alone, with no audio at inference
- **Homophenes:** many phonemes share the same lip shape, so the visual signal is intrinsically ambiguous
- **Data and compute wall:** state-of-the-art systems train on thousands of hours; MuAViC Arabic, German and Russian offer far fewer (19, 13 and 52 hours respectively).
- **Question:** can one parameter-efficient recipe learn to lip-read across languages under these constraints?



What the model sees: 88×88 mouth crops around the lips (MuAViC, Arabic)

The method: a frozen backbone with parameter-efficient adapters

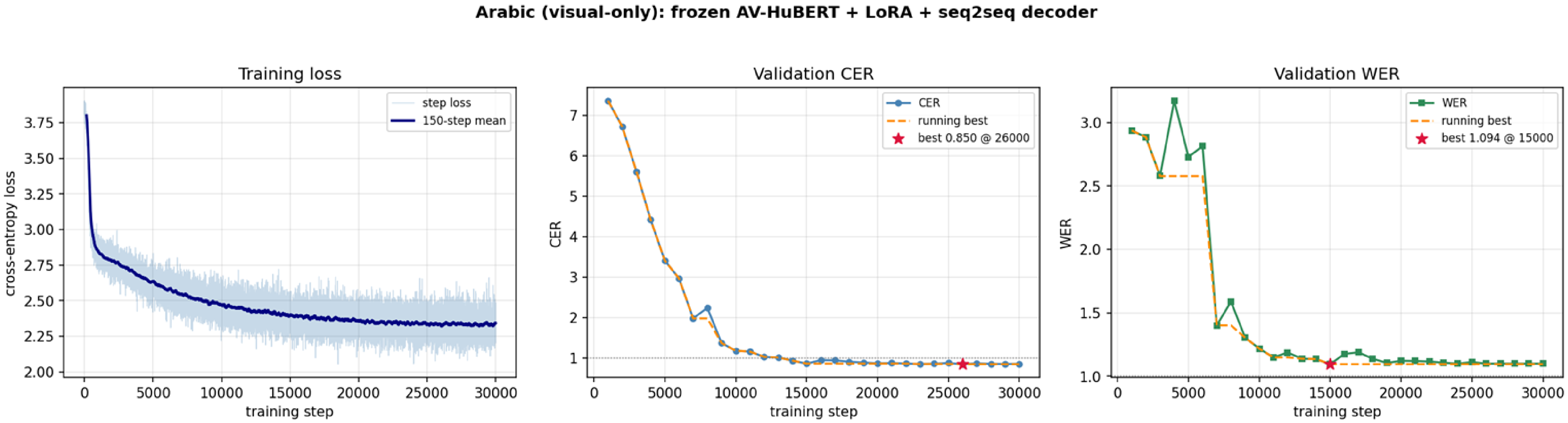
- **Frozen AV-HuBERT (326M)** + LoRA adapters in every attention projection (r=16, 3.1M) + a 4-layer character decoder (17.6M): about 33M trained vs 344M, roughly 10× fewer
- **One recipe, three languages:** applied to Arabic, German and Russian. no per-language tuning



Animated example: a silent MuAViC clip becomes a character-level Arabic transcription

Training objective: CTC vs. sequence-to-sequence

- **CTC (Connectionist Temporal Classification) head collapses:** on frozen features it emits one repeated token regardless of learning rate, rank or head depth, its independence assumption cannot handle ambiguous visual phonemes.
- **Seq2seq decoder restores learning:** validation CER (Character Error Rate) falls from 7.4 to 0.85.



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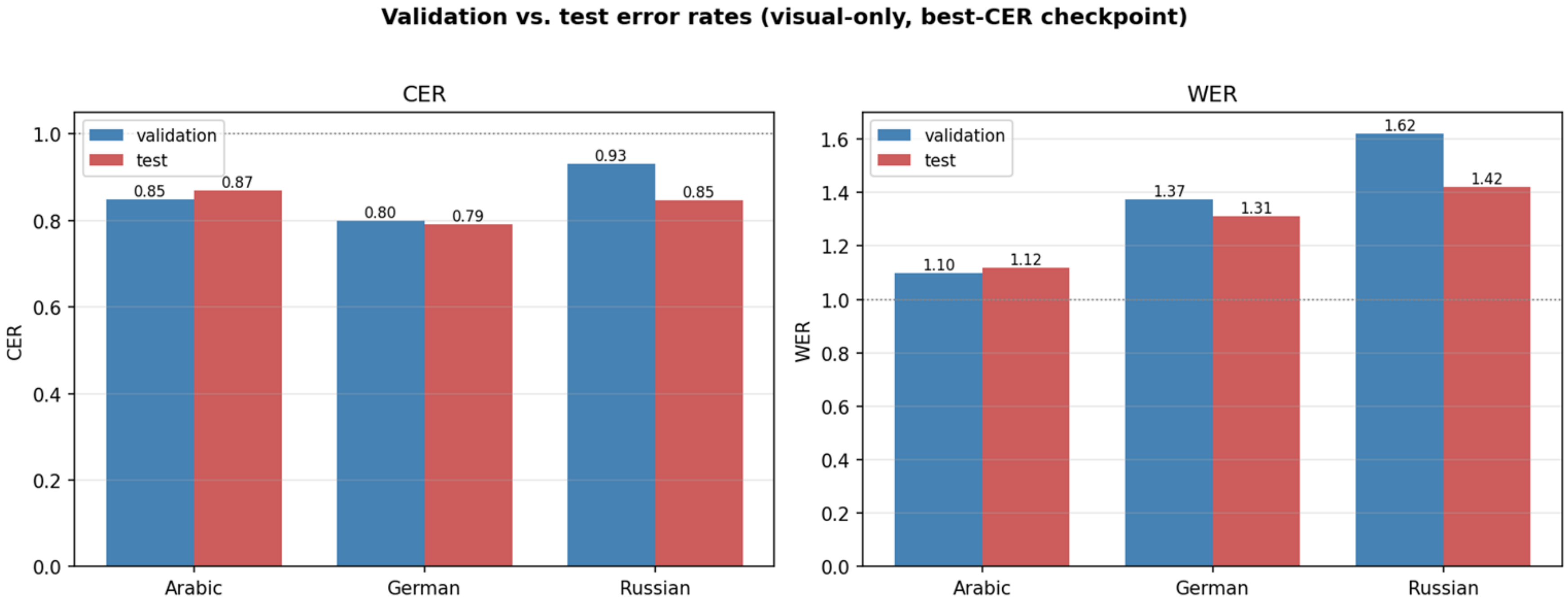
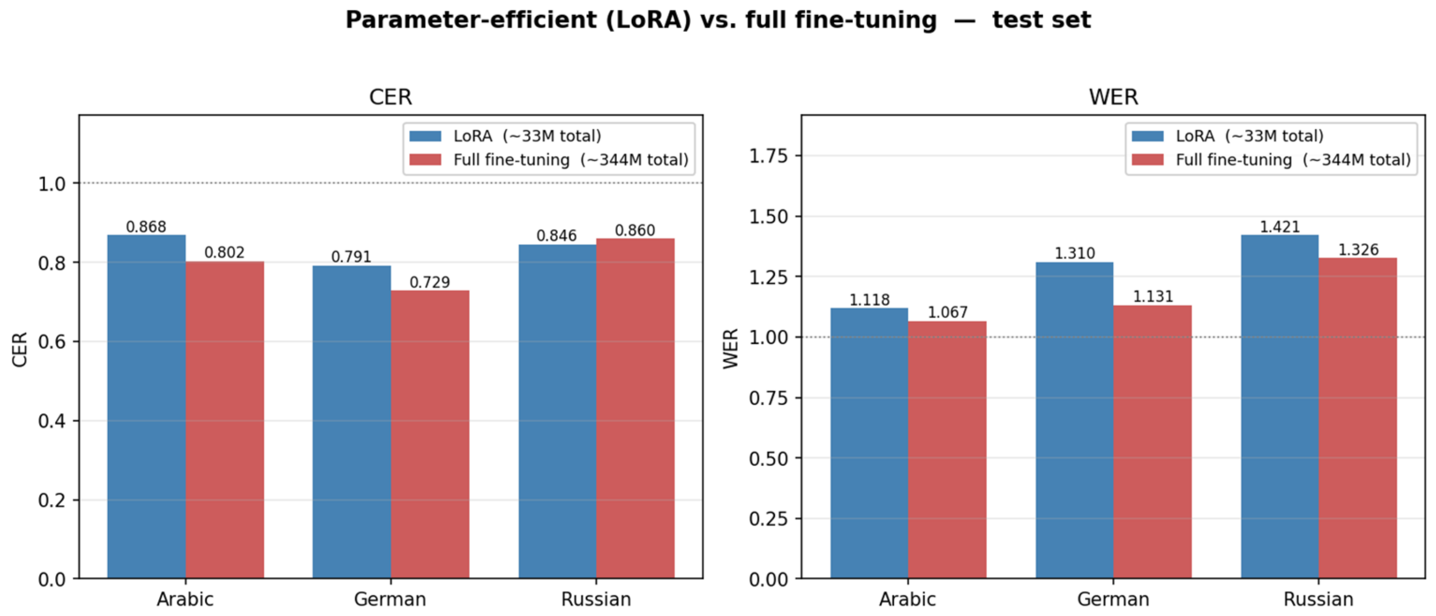
Multilingual evaluation: Arabic, German and Russian

- **No per-language tuning:** identical hyperparameters for Arabic, German and Russian
- **Test CER 0.87 / 0.79 / 0.85:** test tracks validation closely (no overfitting), but absolute error stays high. The next experiments explore why



LoRA vs full fine-tuning (RQ2)

- **Controlled comparison:** same data, decoder and schedule, only the backbone strategy differs (344M trained vs 33M)
- **Result:** full fine-tuning is at most 8% relatively better on test CER (Arabic, German) and LoRA matches it on Russian, at about a tenth of the trained parameters



Data quality: filtering non-face training clips

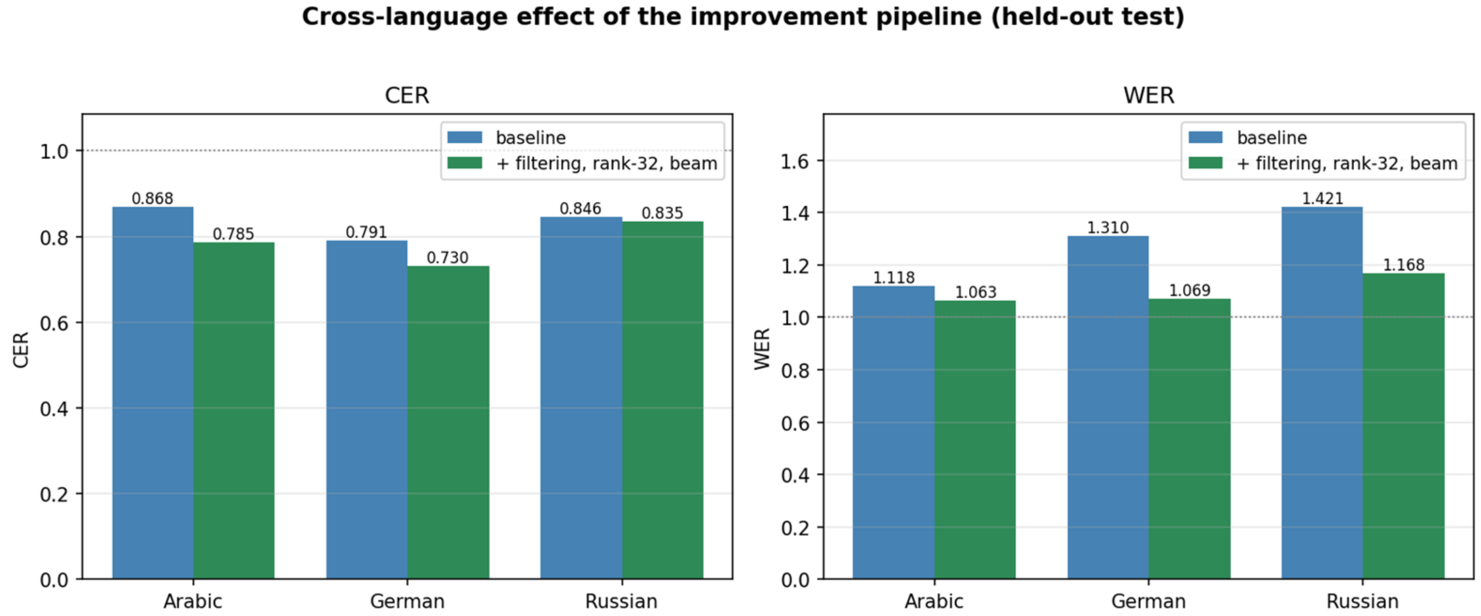
- **Label noise at the source:** MuAViC is cut from TED talks, so title cards, stage shots and audience frames carry transcripts
- **Cheap automatic audit:** per-clip skin fraction is strongly bimodal, a 0.6 threshold removes 16% of the Arabic training clips

Removed: lowest face-fraction clips

Retained: random sample

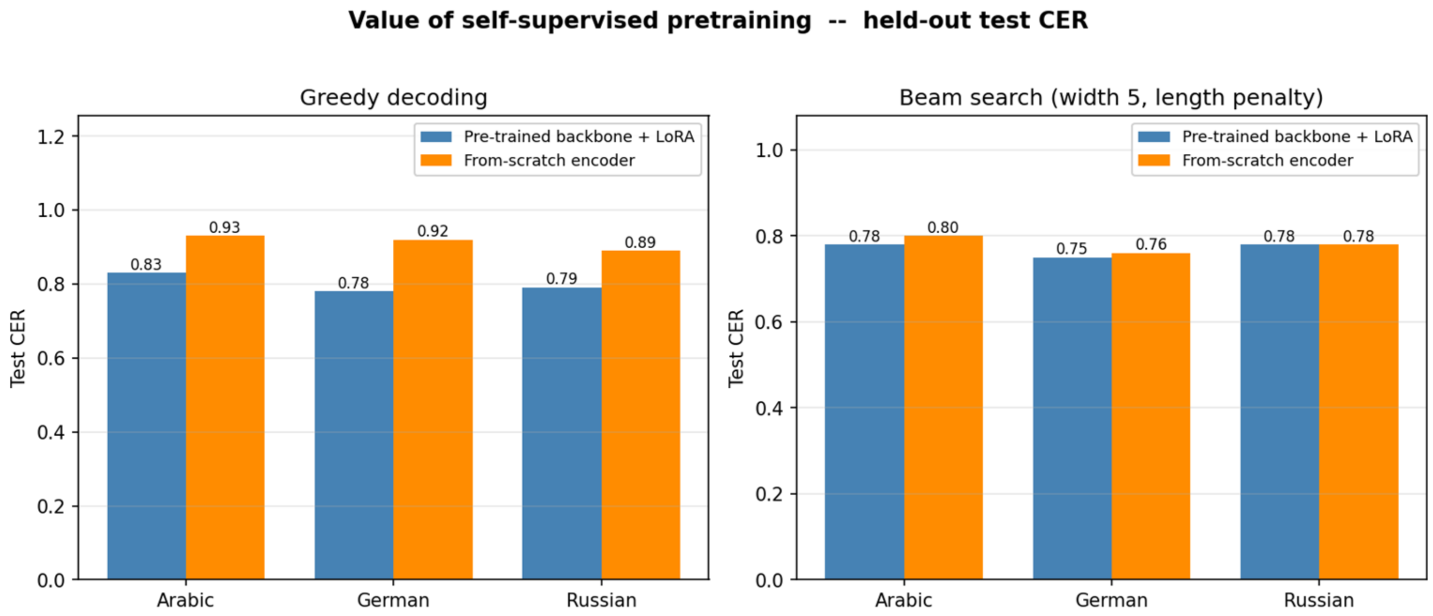
Improvements: data filtering and beam-search decoding

- **Filter, then beam:** retraining on filtered data beats training on more but dirtier data, beam search with a length penalty stops run-on transcripts, the main reason of high WER
- **Arabic test CER 0.87 → 0.83 → 0.78:** about 10% relative improvement at no extra training cost. Similarly, German and Russian improve as well



The performance ceiling: all configurations converge

- **Three independent probes:** full fine-tuning, rank-32 adapters and a from-scratch encoder all land near 0.78 CER / 1.05 WER
- **Why:** outputs look like real valid sentences but are weakly tied to the video input, that is because the visual input is too weak or ambiguous.
- **Pretraining helps convergence, not the ceiling:** the limit is a consequent of the visual-only task



Complete test results

Configuration	Trained	Arabic		German		Russian		Average	
		CER	WER	CER	WER	CER	WER	CER	WER
LoRA, baseline (greedy)	33M	0.87	1.12	0.79	1.31	0.85	1.42	0.84	1.28
LoRA, filtered (greedy)	33M	0.83	1.10	0.78	1.37	0.79	1.18	0.80	1.22
LoRA, filtered + beam	**33M**	**0.78**	**1.07**	**0.75**	**1.12**	**0.78**	**1.02**	**0.77**	**1.07**
Full fine-tuning (greedy)	344M	0.80	1.07	0.73	1.13	0.86	1.33	0.80	1.18
From-scratch encoder + beam	45M	0.80	1.05	0.76	1.09	0.78	0.99	0.78	1.04

Summary of contributions

- **Objective over capacity:** with frozen visual features, seq2seq vs CTC is the deciding factor, not the model size
- **Adapters are enough:** LoRA recovers essentially all of full fine-tuning at about 10× fewer trained parameters
- **Data quality is essential to performance:** removing 16% junk clips improved every language, and beam search prevents run-on sentences
- **A characterized ceiling:** the residual error is set by visual-only, low-resource VSR itself, shown three independent ways

Future work

- **Pre-trained decoder:** start the decoder from a model that already knows the language, so training is spent on grounding text in the video, not on learning spelling, the most promising single change

- **Better input, not bigger models:** extra capacity was shown not to help, so invest in stronger visual feature extractors and more lip-reading pretraining data

- **Broaden the study:** one shared adapter for all languages, and robustness experiments

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Thank you for your attention

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